
Benchmarking Motion Blur Robustness in Modern Object Detectors: A Comparative Study of YOLOv8, RT-DETR, and GroundingDINO

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Abstract

Object detection systems have achieved remarkable performance on benchmark datasets [1-5]; however, their robustness under image degradation remains an important challenge for real-world deployment. Motion blur is a common degradation encountered in surveillance systems [20-23], autonomous vehicles, mobile photography, and robotics, often leading to significant performance deterioration. This paper presents a comparative robustness study of modern object detection architectures under motion blur conditions. We evaluate YOLOv8n, YOLOv8s, RT-DETR, and GroundingDINO on a curated subset of images and corresponding blurred variants. Unlike traditional benchmark studies that primarily focus on mean Average Precision (mAP), this work emphasizes robustness through detection count retention and confidence stability under degradation. Experimental results show substantial performance drops for YOLO-based detectors, while RT-DETR demonstrates higher robustness in this benchmark. GroundingDINO retains partial detection capability but exhibits reduced confidence under severe blur. The findings highlight the importance of robustness evaluation beyond conventional benchmark accuracy and provide insights into architectural behaviour under degraded visual conditions.

Keywords: Object Detection, Motion Blur, YOLOv8, RT-DETR, GroundingDINO, Robustness Analysis, Computer Vision

1. Introduction

Object detection is a fundamental task in computer vision and serves as the foundation for numerous applications, including autonomous driving, intelligent surveillance, robotics, medical imaging, and augmented reality [1,12]. Recent advances in deep learning have significantly improved detection accuracy through architectures such as YOLO, DETR-based detectors, and vision-language grounded models [3,5,9].

Despite impressive benchmark performance, object detectors often operate in environments where image quality is compromised. Real-world images may suffer from motion blur, low illumination, compression artifacts, adverse weather conditions, and sensor noise [17,20,21]. Among these challenges, motion blur remains particularly problematic because it degrades edges, textures, and object boundaries that are critical for detection [22,23].

Most object detection studies focus primarily on benchmark accuracy metrics under ideal conditions [1,5,12]. Comparatively fewer studies investigate robustness under controlled image degradation [17,20]. Understanding how different architectural paradigms respond to blur is important for selecting appropriate models in practical deployments.

This work investigates the robustness of four representative object detection models:

- YOLOv8n
- YOLOv8s
- RT-DETR
- GroundingDINO

The selected models represent three distinct detection paradigms: lightweight CNN-based detection, transformer-based detection, and vision-language grounded detection. Through controlled motion blur experiments, this study aims to provide insights into architectural strengths and weaknesses under degraded visual conditions.

The main contributions of this work are:

1. A comparative robustness benchmark across CNN-based, transformer-based, and vision-language object detectors.
2. Quantitative analysis of detection count degradation and confidence score degradation under motion blur.
3. Qualitative visualization of model behaviour under degraded conditions.
4. Practical insights into model suitability for real-world deployment scenarios involving image degradation.

2. Related Work

2.1 Object Detection

Object detection has evolved significantly over the past decade. Early deep learning approaches such as R-CNN and Faster R-CNN demonstrated substantial improvements over traditional methods. Subsequently, single-stage detectors such as YOLO achieved real-time performance while maintaining competitive accuracy [1,2,3].

YOLOv8 continues this progression by introducing architectural improvements that enhance detection efficiency and inference speed. The lightweight YOLOv8n variant prioritizes computational efficiency, while YOLOv8s offers improved detection performance through increased model capacity.

2.2 Transformer-Based Detection

Transformer architectures have transformed many areas of computer vision [15,16]. DETR introduced a set-prediction framework that eliminates hand-crafted components such as anchor generation and non-maximum suppression [5].

RT-DETR extends this paradigm by combining transformer-based reasoning with real-time performance characteristics [8]. The model leverages global attention mechanisms that may provide increased robustness under degraded image conditions.

2.3 Vision-Language Detection

Recent developments have expanded object detection beyond closed-set categories. GroundingDINO integrates visual and textual information to perform open-vocabulary object detection [9]. By aligning image features with language representations, GroundingDINO can detect objects described through text prompts [9,11].

Although these capabilities offer significant flexibility, the robustness of vision-language alignment under image degradation remains an active research area [9].

2.4 Robustness Under Image Degradation

Robustness evaluation has gained increasing attention due to the gap between benchmark performance and real-world deployment conditions [17,18,19,20]. Previous studies have investigated the effects of noise, weather conditions, adversarial perturbations, and blur [20-23]. However, comparative analyses spanning CNN-based, transformer-based, and vision-language detectors remain relatively limited.

This study contributes to this area by examining motion blur robustness across multiple architectural families.

3. Methodology

3.1 Dataset

We evaluate YOLOv8n, YOLOv8s, RT-DETR-L, and GroundingDINO on a benchmark consisting of 100 COCO validation images and corresponding motion-blurred variants [12]. The selected images contain diverse object categories, scene compositions, and object scales, enabling robustness evaluation across multiple visual contexts. The same image subset was used for all evaluated models to ensure a fair comparison.

3.2 Motion Blur Generation

Motion blur was synthetically applied using OpenCV-based motion blur kernels. To better approximate real-world camera and object motion, blur direction was randomly varied for each image rather than being restricted to horizontal or vertical motion. Multiple blur intensities were initially explored using kernel sizes of 15, 25, and 45 pixels. Final evaluation was conducted using the heavy blur setting (kernel

size = 45), representing a challenging degradation scenario. Identical blur-generation procedures were applied across all evaluated models. The blur generation process simulates realistic image degradation encountered in mobile photography, surveillance systems, and moving cameras [22,23].

3.3 Evaluated Models

YOLOv8n

YOLOv8n is the smallest variant of the YOLOv8 family and prioritizes computational efficiency while maintaining reasonable detection performance [31].

YOLOv8s

YOLOv8s provides increased model capacity and improved feature representation compared to YOLOv8n [31].

RT-DETR

RT-DETR is a transformer-based object detector that combines global attention mechanisms with real-time inference capabilities [8].

GroundingDINO

GroundingDINO is a vision-language grounded detector capable of open-vocabulary object detection through textual prompts [9].

3.4 Evaluation Metrics

The evaluation focuses on relative performance changes under motion blur, providing a robustness-oriented perspective complementary to conventional localization-based metrics. Traditional mAP evaluation requires complete ground-truth matching and primarily measures localization accuracy. Since the focus of this study is robustness under controlled degradation, alternative metrics were adopted:

Detection Count

The average number of detected objects per image.

Average Confidence

The mean confidence score assigned to detections.

Detection Retention

The proportion of detections preserved after blur degradation

Equation:

$$\text{DRR (\%)} = (\text{Average Detections on Blurred Images} / \text{Average Detections on Original Images}) \times 100$$

Confidence Retention

The proportion of confidence preserved after blur degradation.

Equation:

$$\text{CRR (\%)} = (\text{Average Confidence on Blurred Images} / \text{Average Confidence on Original Images}) \times 100$$

4. Experimental Setup

Experiments were conducted using Python-based inference pipelines. Pretrained weights were used for all models without additional fine-tuning.

The evaluation dataset consisted of 100 original images and their corresponding motion-blurred counterparts, resulting in 200 evaluated images in total.

The same original and blurred image sets were evaluated across all models. Detection thresholds were selected according to model-specific recommendations while maintaining consistency throughout the evaluation process.

Qualitative comparisons were performed using a representative umbrella-scene image to analyze model behaviour under motion blur degradation.

5. Results

5.1 Quantitative Results

Detection Performance

Model	Original Detections	Blurred Detections
YOLOv8n	5.65	1.31
YOLOv8s	6.58	1.51
RT-DETR	14.82	6.87
GroundingDINO	6.94	2.71

Confidence Performance

Model	Original confidence	Blurred Confidence
YOLOv8n	0.6079	0.3239
YOLOv8s	0.6521	0.3680
RT-DETR	0.5982	0.4656
GroundingDINO	0.4708	0.4031

Detection Retention Rate (DRR)

Model	Retention Rate
YOLOv8n	23.2%
YOLOv8s	22.9%
RT-DETR	46.4%
GroundingDINO	39.0%

Confidence Retention Rate (CRR)

Model	Confidence Rate
YOLOv8n	53.3%
YOLOv8s	56.4%
RT-DETR	77.8%
GroundingDINO	85.6%

The observed robustness trends were consistent with the aggregated evaluation metrics, indicating notable differences among the evaluated architectures.

Figure 1: Qualitative comparison of object detection performance under severe motion blur. The first image shows the motion-blurred input sample. The remaining panels illustrate predictions generated by YOLOv8n, YOLOv8s, RT-DETR, and GroundingDINO, respectively. RT-DETR preserves the largest number of object detections under degradation, while YOLO-based models exhibit substantial detection loss. GroundingDINO retains partial object understanding despite reduced confidence.



Original Image



Sample image

YOLOv8n - img_67.jpg



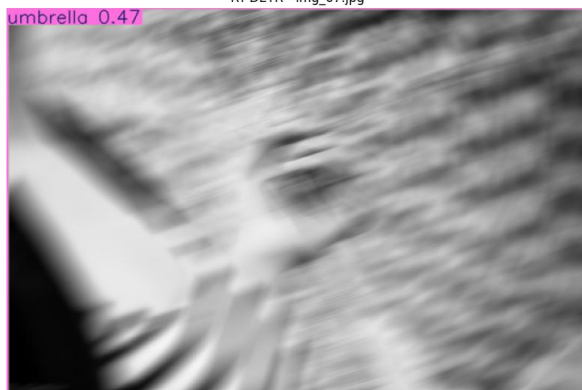
YOLOv8n prediction

YOLOv8s - img_67.jpg



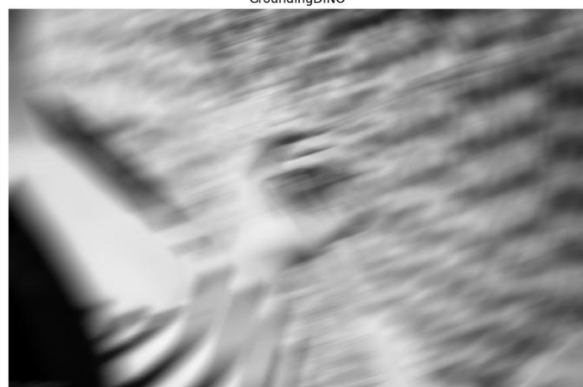
YOLOv8s prediction

RT-DETR - img_67.jpg



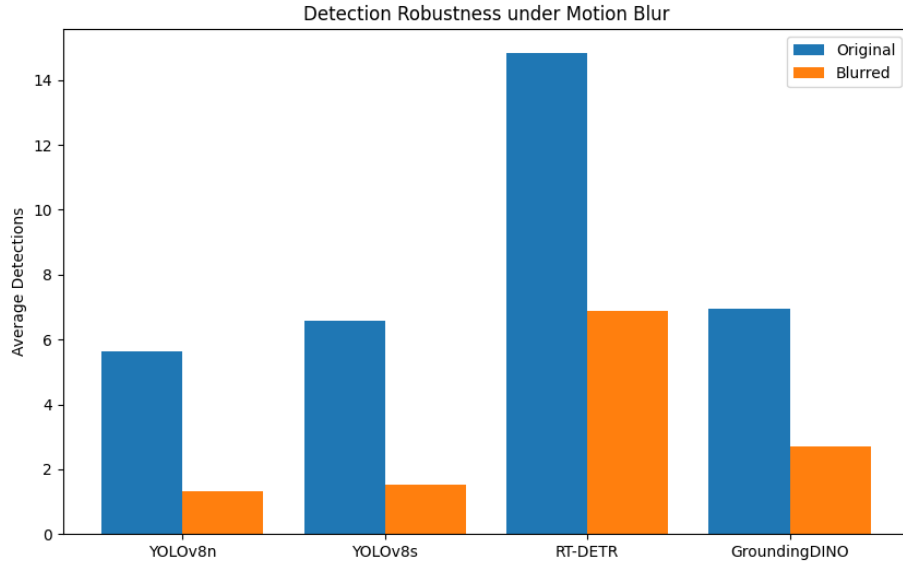
RT-DETR prediction

GroundingDINO



GroundingDINO prediction

Figure 2: Quantitative comparison of object detector robustness under severe motion blur. Bars represent average detection counts on original and blurred images. RT-DETR demonstrates the highest detection retention rate, whereas YOLOv8n and YOLOv8s experience the largest performance degradation. GroundingDINO exhibits intermediate robustness between transformer-based and CNN-based detectors.



5.2 Qualitative Analysis

Qualitative evaluation reveals clear architectural differences. Under severe blur, YOLO-based detectors frequently miss objects that are successfully detected in the original image. RT-DETR maintains stronger localization performance and continues identifying major objects despite degraded visual information.

GroundingDINO exhibits an interesting behaviour pattern. Although confidence values decrease, the model often preserves partial object understanding through text-image alignment. However, severe blur weakens this alignment and reduces detection reliability.

6. Discussion

The experimental results demonstrate that architectural design significantly influences robustness to motion blur.

YOLO models rely heavily on local texture and edge information. Motion blur suppresses these features, leading to substantial degradation in detection performance.

RT-DETR demonstrates the strongest robustness among evaluated models in this benchmark. The global reasoning capabilities enabled by transformer attention mechanisms appear to improve resilience against feature degradation.

GroundingDINO occupies an intermediate position. The model benefits from language-guided reasoning but remains vulnerable when visual information becomes severely degraded.

These observations suggest that future object detection systems may benefit from combining transformer-based reasoning with multimodal representations to improve robustness under challenging conditions.

7. Limitations

Several limitations should be acknowledged. The study was conducted on a subset of 100 COCO validation images, and larger-scale evaluations may provide additional insights. The analysis focused on robustness-oriented metrics (detection retention and confidence retention) rather than localization-based metrics such as mAP and IoU. Furthermore, only motion blur was investigated, while other real-world degradations remain unexplored. Statistical variability and significance testing were not performed because only aggregated metrics were retained; therefore, the results should be interpreted as descriptive robustness indicators rather than definitive performance rankings. Confidence scores are model-dependent; therefore cross-model confidence comparisons should be interpreted with caution.

Future work will extend the benchmark to larger datasets, additional degradations such as Gaussian noise, low-light conditions, rain, fog, and compression artifacts and complementary evaluation metrics.

8. Conclusion

This paper presented a comparative robustness benchmark of YOLOv8n, YOLOv8s, RT-DETR, and GroundingDINO under motion blur degradation. The results demonstrate that motion blur significantly affects object detection performance across all evaluated models. RT-DETR exhibited the strongest robustness, maintaining substantially higher detection capability under degraded conditions.

GroundingDINO retained partial detection ability but experienced confidence degradation. YOLO-based models showed the largest performance decline.

The findings highlight the importance of robustness evaluation beyond conventional benchmark accuracy and provide practical guidance for selecting object detection models in real-world environments where image degradation is unavoidable.

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Appendix A: Experimental Details

A.1 Hardware and Software Setup

All experiments were conducted using Google Colab with GPU acceleration.

Hardware

- NVIDIA Tesla T4 GPU
- High-RAM Google Colab environment

Software Environment

- Python 3.11
- PyTorch: 2.11.0+cu128
- CUDA: 12.8
- Transformers 5.0.0
- Ultralytics: 8.4.60
- OpenCV 4.13.0

- NumPy 2.0.2
- Pandas 2.2.2
- Matplotlib
- Pillow

Models Evaluated

- YOLOv8n
- YOLOv8s
- RT-DETR-L
- GroundingDINO

All models used publicly available pretrained weights without additional fine-tuning.

A.2 Motion Blur Generation Parameters

Blurred images were generated from original images using OpenCV-based blur operations.

To simulate realistic motion degradation, blur orientation was randomized during kernel generation rather than being restricted to fixed horizontal or vertical directions. This approach better reflects practical scenarios involving camera shake, object motion, and handheld image acquisition.

Three blur intensities were initially explored:

Blur Level	Kernel Size
Light	15
Medium	25
Heavy	45

Preliminary experiments were conducted using kernel sizes of 15 and 25 pixels; however, only the severe blur setting (kernel size = 45) is reported in this paper because it produced the most pronounced robustness differences among models.

Blurred images were generated using identical parameters across all evaluated models to ensure fair comparison.

A.3 GroundingDINO Prompt

The following prompt was used during GroundingDINO inference:

person. bicycle. car. motorcycle. airplane. bus. train. truck. boat. traffic light. fire hydrant. stop sign. bench. bird. cat. dog. horse. sheep. cow. elephant. bear. zebra. giraffe. backpack. umbrella. handbag.

suitcase. bottle. cup. chair. couch. potted plant. bed. dining table. laptop. mouse. keyboard. cell phone. book.

The prompt was selected to cover common object categories present within the evaluation images.

A.4 Evaluation Metrics

The following metrics were used:

Detection Count (DC)

Average number of objects detected per image.

Average Confidence (AC)

Average confidence score assigned to detections.

Detection Retention Rate (DRR)

Percentage of detections retained after blur degradation.

Confidence Retention Rate (CRR)

Percentage of confidence retained after blur degradation.

These metrics were selected to measure robustness trends under controlled image degradation.